Workout 1A Homework 1 Challenge

Below is a scored review of your assessment. All questions are shown.

Earned 69,993 of 99,990 points. 70%

Main Question Set

Solve six of the first ten questions to achieve a passing score of 60%. Don't miss the learning opportunity in the rest of the problems, though -- spend the full time and attempt every one of them!

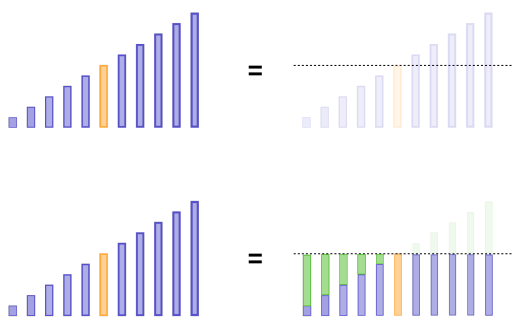
Correct Answer Partially Correct Incorrect Answer

1 A set of 27 numbers which are spaced evenly apart have a sum of 108. What is the value of the number that is in the middle?

Your Answer	4	
Correct Answer	4	

Explanation

It's always true that in a sequence of evenly spaced numbers (where the amount added to any term in order to get the next term is always the same amount), the very middle number is the same as the average of the whole set of numbers. The diagram below illustrates this with 11 towers which represent 11 evenly-spaced numbers. Which color represents the middle number, the average?



If you said "yellow," then you're right! This is the sixth number from the left. The average can be represented by the height of the towers on the right, which were obtained by chopping off bits of the taller ones and adding them to the shorter ones (the green pieces). The average is simply the value of the numbers (the height of the towers) if you pretend that all of the numbers were the same.

Since we want the very middle number of a set of 27 equally-spaced numbers, we just want the average of all the numbers. The answer is then the following:

$$\text{middle number } = \text{average} = \frac{108}{27} = \boxed{4.}$$

2

The sum of the first 2 odd numbers is equal to $1+3=4$.

The sum of the first 3 odd numbers is equal to $1+3+5=9$.

The sum of the first 4 odd numbers is equal to $1+3+5+7=16$.

Following this pattern, the sum of the first 10 odd numbers is equal to which of the following?

Your Answer	10^2
Correct Answer	10^2

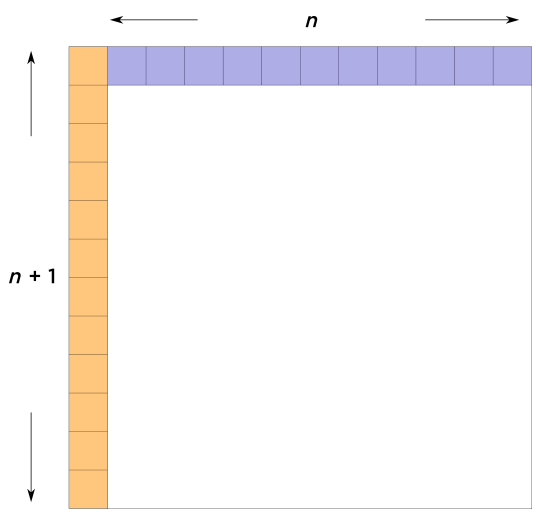
Explanation

The curious pattern to the sum of the first n odd numbers is this:

$$(1^{\text{st}} \text{ odd}) + (2^{\text{nd}} \text{ odd}) + (3^{\text{rd}} \text{ odd}) + \ldots + (n^{\text{th}} \text{ odd}) = n^2.$$

This question is asking the same thing as, "What is the sum of the numbers $1+3+5+7+\ldots+19$?", except we're even better off, since we don't even have to figure out what n equals -- we are given that n equals 10. In fact, you don't even have to know that the 10th odd is equal to 19. The sum of the first 10 odds is equal to $\boxed{10^2}$.

Math note: It's quite easy to see why this formula works. It's because in order to go from one square, n^2, to the next consecutive square, $(n+1)^2$, you simply add $n+n+1$, which is always an odd number. We start from 1, which is a square, and adding consecutive squares will always result in another square number.



3

In shoe sizing, a given number shoe size for a women's shoe is actually a different size from the same number size for a men's shoe. The chart below shows a size conversion between women's and men's shoes. Let w equal a women's shoe size, and m the equivalent size in men's shoes. What is the relationship between w and m?

Women's Sizing	7.5	8	8.5	9	9.5
Men's Sizing	6	6.5	7	7.5	8

Your Answer $w = 1.5 + m$

Correct Answer $w = 1.5 + m$

Explanation

Let's look at a sample point. An 8.5-size women's shoe is equivalent to a 7-size men's shoe. The woman's shoe size is 1.5 larger than the men's shoe size. Checking the other points confirms this pattern, so the answer is $\boxed{\text{choice i.), } w = 1.5 + m}$.

4 Back in the time of the Romans, shoes were originally measured in terms of a unit called barley corns. Instead of barley corns, we now define a unit of shoe size to be 1 centimeter in every dimension. What percent is the volume of a size 8.5 foot 24.5 cm long compared with a size 9 foot measuring 25 cm long? Choose the closest answer.

Your Answer	94.4%
Correct Answer	94.4%

Explanation

The ratio of width of the size-8.5 shoe compared with the size-9 shoe is $\frac{24.5}{25}$, which, multiplying by top and bottom, is equal to $\frac{49}{50}$. The length ratio and height ratio are similarly also $\frac{49}{50}$. Since volume multiplies by product of the width, length and height factors, the volume of the size-8.5 shoe will be $\left(\frac{49}{50}\right)^3 = \frac{49^3}{125000} = \frac{117649}{1250000} \approx \frac{118}{125}$. Since we want to find the closest answer, and our answer choices are each different by at least 1%, we can round off this answer to the third digit, which only causes an error of at most 1%. Conveniently, the denominator of $\frac{118}{125}$ lends itself to being multiplied by 8 to give a 1000 in the bottom, which will allow us to calculate the percent easily. Thus the fraction becomes $\left(\frac{118}{125}\right)\left(\frac{8}{8}\right) = \frac{944}{1000} = 0.944 \rightarrow \boxed{94.4\%}$.

5 For Halloween, a pastry chef bakes pumpkin cookies and skeleton cookies. The pumpkin cookies are 3 inches long and 3.5 inches wide. The skeleton cookies are 4 inches long and 1.5 inches wide. If the pastry chef was able to fit 96 pumpkin cookies on his cookie sheets, how many skeleton cookies can he fit on all of his cookie sheets? Your answer should be the number only, without units.

✓	**Your Answer**	168
	Correct Answer	168

Explanation

Each pumpkin cookie has area $3 \times 3.5 = 10.5 \text{ in}^2$ compared to the area of a skeleton cookie, $4 \times 1.5 = 6 \text{ in}^2$, so intuitively we should be able to fit more skeleton cookies than pumpkin cookies on one sheet. We want to think about the proportion of space taken up by a skeleton cookie compared with a pumpkin cookie. The proportion of skeleton cookies is this same ratio, but flipped.

The skeleton cookies are 4 inches long, compared with the pumpkin cookies, which are 3 inches long, so in one sheet, we can only fit $\frac{3}{4}$ the number of rows of skeleton cookies as pumpkin cookies. height space of the pumpkin cookies. The skeleton cookies are 1.5 inches wide compared with the pumpkin cookies, which are 3.5 inches wide, so we can fit $\frac{3.5}{1.5} = \frac{7}{3}$ as many vertical columns of skeleton cookies as pumpkin cookies. Multiplying these together, we get $\frac{3}{4} \times \frac{7}{3} = \frac{7}{4}$ as many skeleton cookies as pumpkin cookies. We had 96 pumpkin cookies, so we should be able to fit $96\left(\frac{7}{4}\right) = 24 \times 7 = \boxed{168}$ skeleton cookie.

6 Suppose you started with a single-spaced 6-page report in single-spaced 8-point font and switched to 12-point double-spaced font. How many pages will your report end up being?

Your Answer	18 pages
Correct Answer	27 pages

Explanation

Compare the dimensions of one letter in 8-point and also in 12-point. The 12-point letter will have $\frac{3}{2}$ the width and $\frac{3}{2}$ the height of the same letter in 8-point. The overall space the 12-point font takes up will be $\left(\frac{3}{2}\right)^2 = \frac{9}{4}$ times the space. Making the report double-spaced will multiply the space used by 2, so the overall space multiplies by $2\left(\frac{9}{4}\right) = \frac{9}{2}$. Thus the 6 pages will turn into $6\left(\frac{9}{2}\right) = \boxed{27}$ pages.

7 Evan averages 88 points on his five tests. The average of his two top scores is 92, and the average of his two low scores is 85. What was his middle score?

Your Answer	86
Correct Answer	86

Explanation

The average of his scores is 88, so we could pretend that each exam score is 88. We now want to adjust the scores by moving points around, without creating a net difference in the total points. For example, we can add 4 points to the top two scores to make them 92, but we'd have to remove them from somewhere else. Removing 4 each from the two low scores makes each low score 84. However, the two low scores are supposed to have an average of 85, so we must steal 2 points from the middle score, one for each of the two lower exam scores. Now the middle score is $88 - 2 = \boxed{86}$, and this is the answer.

8 On a distant planet, two-thirds of the aliens have six legs and the rest have four legs. If there are 2560 legs on the planet, then how many aliens are on the planet?

Your Answer	480
Correct Answer	480

Explanation

Suppose there were one 4-legged alien and two 6-legged aliens on the planet. Combined, they would have $4 + 2(6) = 16$ legs. This comprises one subgroup of the creatures on this planet. We just need to scale the subgroup in order to get 2560 legs. We recognize 256 as being 16^2, so the planet has 160 of these groups of 3 aliens, giving $3 \times 160 = \boxed{480}$ total aliens.

9 Sam eats twice as many gummy bears as Vishanth, and one-third as many gummy bears as Pei. If Pei eats 90 more gummy bears than Vishanth, then how many gummy bears were eaten altogether?

Your Answer	[no answer given]
Correct Answer	162

Explanation

Let's create a variable for the number of gummy bears that Vishanth eats and relate the other people's gummy bear amounts to Vishanth's amount. Since Sam eats one-third Pei's amount, Pei eats three times as much as Sam eats.

$$\begin{aligned} x &= (\text{Number of gummy bears Vishanth eats}) \\ 2x &= (\text{Number of gummy bears Sam eats}) \\ 3(2x) = 6x &= (\text{Number of gummy bears Pei eats}) \end{aligned}$$

The difference in the number of gummy bears Vishanth ate and Pei ate is equal to $5x$, which is given to equal 90 gummy bears. Thus $x = \frac{90}{5} = 18$. The total number of gummy bears eaten equals $x + 2x + 6x = 9x = 9(18) = \boxed{162}$.

10 What is $\frac{17+19+21+23+25+\ldots+151+153}{15}$?

✕	**Your Answer**	[no answer given]
	Correct Answer	391

Explanation

We saw in the lesson that for a sequence with equal spacing between numbers (called an arithmetic sequence), we can find the sum of all of the terms by multiplying the number of terms with the average of the terms. To find the number of numbers, find the difference between the first and last terms and divide this by 2, since the gap between each number is 2. This gives us the number of gaps, $\frac{153-17}{2} = \frac{136}{2} = 68$. The number of numbers is one larger than the number of gaps, so there are 69 numbers. Now let's find the average number in the sequence. This is equal to the average of the first and last terms, $\frac{17+153}{2} = \frac{170}{2} = 85$. Putting this together, the sum of the sequence divided by 15 equals:

$$\begin{aligned}
\frac{1}{15} \times (\text{Average number}) \times (\#\text{ of numbers}) &= \frac{85 \times 69}{15} \\
&= \frac{5 \times 17 \times 23 \times 3}{15} \\
&= (20+3)(20-3) \qquad (\text{Diff. of squares}) \\
&= 400 - 9 \\
&= \boxed{391}.
\end{aligned}$$

Earned 0 of 7 points. 0%

Bonus Challenge Questions

These optional questions will not count towards your final score. However, we strongly suggest trying them for the benefit of the challenge!

11 The sum of 4 consecutive odd integers is 56. What is the smallest number?

Your Answer	[no answer given]
Correct Answer	11

Explanation

The average of the four numbers is $\frac{56}{4} = 14$. Thus the middle two numbers average to 14 also, so they must be 13 and 15. Since the numbers are consecutive odds, we then know that the four numbers are $11, 13, 15,$ and 17, so the smallest number is $\boxed{11}$.

12

Compare $\frac{\sqrt{24}}{\sqrt{5}}$ and $\frac{\sqrt{75}}{\sqrt{18}}$.

Which one is bigger?

Your Answer [no answer given]

Correct Answer $\frac{\sqrt{24}}{\sqrt{5}}$

Explanation

First, simplify the fractions by removing square factors from inside the square root.

$$\frac{\sqrt{24}}{\sqrt{5}} \text{ vs. } \frac{\sqrt{75}}{\sqrt{18}}$$

$$\frac{2\sqrt{6}}{\sqrt{5}} \text{ vs. } \frac{5\sqrt{3}}{3\sqrt{2}}$$

$$\frac{2\sqrt{2}}{\sqrt{5}} \text{ vs. } \frac{5}{3\sqrt{2}} \qquad (\text{Divide both sides by} \sqrt{3})$$

$$\frac{2\sqrt{2}}{\sqrt{5}}\left(\frac{3\sqrt{2}}{3\sqrt{2}}\right) \text{ vs. } \frac{5}{3\sqrt{2}}\left(\frac{\sqrt{5}}{\sqrt{5}}\right) \qquad (\text{Make common denominator}$$

$$2\sqrt{2} \times 3\sqrt{2} \text{ vs. } 5\sqrt{5}$$

$$12 \text{ vs. } 5\sqrt{5}.$$

Now let's estimate $\sqrt{5}$. Since 5 is larger than 2^2 and smaller than 3^2, $\sqrt{5}$ is between 2 and 3, closer to 2. We'll have to esimate more precisely. Squaring both numbers gives:

$$12^2 \text{ vs. } (5\sqrt{5})^2$$
$$144 \text{ vs. } 125$$

Thus $\boxed{\frac{\sqrt{24}}{\sqrt{5}}}$ is larger.

13 Approximately how many digits are there in the number $\frac{7^{77}}{5^{37}}$?

1. digits

Your Answer	1. [no answer given]
Correct Answer	1. 40

Explanation

Use the fact that $7^2 = 49 \approx 50 = 5 \times 10$. Then $7^{2n} \approx 5^n \times 10^n$. We can write:

$$\begin{aligned}\frac{7^{77}}{5^{37}} &= \frac{7(7^{76})}{5^{37}} \\ &\approx \frac{7(5^{38}10^{38})}{5^{37}} \\ &= 7(5)10^{38} \\ &= 35 \times 10^{38}.\end{aligned}$$

This number equals 35 followed by 38 zeroes, so it has 40 digits.

14 Suppose that in 4 minutes, 6 experienced kids can capture 3 flags and 13 inexperienced kids can capture 4 flags. How many minutes will it take 2 inexperienced and 3 experienced kids to capture 55 flags?

Your Answer	[no answer given]
Correct Answer	104

Explanation

First find the flag-capture rate for one experienced kid and also for one inexperienced kid. The following shows the computation for one experienced kid, using the idea that the time and number of kids is inversely related, but the number of flags is directly related to the number of kids as well as the number of minutes. In other words, if the number of flags multiplies by x, the number of minutes multiplies by x or the number of kids needed multiplies by x, but if the number of kids multiplies by x, the number of minutes decreases by a factor of $\frac{1}{x}$.

$$\begin{array}{lll} 6 \text{ experienced kids} & \text{capture } 3 \text{ flags} & \text{in } 4 \text{ minutes} \\ 1 \text{ experienced kid} & \text{captures } 3 \text{ flags} & \text{in } 24 \text{ minutes} \\ 1 \text{ experienced kid} & \text{captures } 1 \text{ flag} & \text{in } 8 \text{ minutes} \end{array}$$

Similarly, we can do the same thing to find the unit "rate" for one inexperienced kid.

$$\begin{array}{lll} 13 \text{ inexperienced kids} & \text{capture } 4 \text{ flags} & \text{in } 4 \text{ minutes} \\ 1 \text{ inexperienced kid} & \text{captures } 4 \text{ flags} & \text{in } 52 \text{ minutes} \\ 1 \text{ inexperienced kid} & \text{captures } 1 \text{ flag} & \text{in } 13 \text{ minutes} \end{array}$$

The speed at which one experienced kid can capture flags is $\frac{1}{8}$ flag per minute, and the speed for one inexperienced kid is $\frac{1}{13}$ flag per minute. If we have 3 experienced kids and 2 experienced kids, their combined rate of working will be equal to the following:

$$\begin{aligned} \text{Combined rate} &= \frac{3}{8} + \frac{2}{13} \\ &= \left(\frac{3}{8}\right)\left(\frac{13}{13}\right) + \left(\frac{2}{13}\right)\left(\frac{8}{8}\right) \\ &= \frac{39+16}{8 \times 13} \\ &= \frac{55}{104} \text{ flags per minute .} \end{aligned}$$

Conveniently, the problem asks us to capture 55 flags, so it will take $\boxed{104}$ minutes to do this.

15 How many different solutions (m, n) are there that satisfy $2^m = 9^n$ for some integers m and n?

Your Answer	[no answer given]
Correct Answer	1

Explanation

The number 2 is even, so since if m is greater than 0, 2^m is even as well. On the other hand, 9^n is always odd. The only possible value for m is 0, which makes $n = 0$ as well, giving us $2^0 = 1 = 9^0$. The answer is $\boxed{1}$.

16 What is the value of $7(2^0) + 6(2^1) + 5(2^2) + 4(2^3) + 3(2^4) + 2(2^5) + 2^6$?

Your Answer	[no answer given]
Correct Answer	247

Explanation

Notice that we can create this sum by adding 7 sub-sequences, each progressively shorter and shorter.

$$\begin{aligned}
2^6 + 2^5 + 2^4 + 2^3 + 2^2 + 2^1 + 2^0 &= 2^7 - 1 \\
+2^5 + 2^4 + 2^3 + 2^2 + 2^1 + 2^0 &= 2^6 - 1 \\
+2^4 + 2^3 + 2^2 + 2^1 + 2^0 &= 2^5 - 1 \\
+2^3 + 2^2 + 2^1 + 2^0 &= 2^4 - 1 \\
+2^2 + 2^1 + 2^0 &= 2^3 - 1 \\
+2^1 + 2^0 &= 2^2 - 1 \\
+2^0 &= 2^1 - 1
\end{aligned}$$

Adding the right-hand sides, we see that we get another sum of powers of 2.

$$\begin{aligned}
(2^1 - 1) + (2^2 - 1) + (2^3 - 1) + (2^4 - 1) + (2^5 - 1) + (2^6 - 1) + (2^7 - 1) &= (2^0 + 2^1 + 2^2 + \ldots + 2^7) - 7 - 2^0 \\
&= 2^8 - 1 - 7 - 1 \\
&= 256 - 9 \\
&= \boxed{247}.
\end{aligned}$$

17 If $\dfrac{2020^2 - a^2}{2017^2 - b^2} = 7$ and a is an integer, what is the value of a?

✕	**Your Answer**	[no answer given]
	Correct Answer	2013

Explanation

In the Day 1 Simple Simplification lesson, recall we were asked to simplify $\frac{2020^2-2012^2}{2017^2-2015^2}$. The numbers were nice: $2020 + 2012$ and $2017 + 2015$ were equal, so using difference of squares $(a^2 - b^2 = (a+b)(a-b))$, we were able to cancel terms to make this reduce down to $\frac{2020-2012}{2017-2015} = \frac{8}{2} = 4$.

Solution 1. Use the same idea as in the Day 1 lesson, but in reverse. Difference of squares will give us

$$\frac{2020^2 - a^2}{2017^2 - b^2} = \frac{(2020+a)(2020-a)}{(2017+b)(2017-b)}.$$

We hope to make some of these factors cancel, by perhaps setting $2020 + a$ equal to $2017 + b$.

We would like this to equal 1

$$\frac{(2020+a)(2020-a)}{(2017+b)(2017-b)} = 7$$

If a and b are too small, then $\frac{2020-a}{2017-b}$ will not be an integer, so we want a and b to be large. Since we cancel $\frac{2020+a}{2017+b}$, then we know $\frac{2020-a}{2017-b}$ must equal 7.

a and *b* should be large

$$\frac{(2020+a)(2020-a)}{(2017+b)(2017-b)} = 7$$

this should equal 7

For $2020 + a = 2017 + b$, then $b = a + 3$. The largest possible value of b which makes $2017 - b$ non-zero is $b = 2016$, which would make the largest possible value of a equal to 2013. Plugging these values in, we get

$$\frac{(2020+2013)(2020-2013)}{(2017+2016)(2017-2016)} = \frac{(4033)(7)}{(4033)(1)}$$

$$= 7.$$

Solution 2.

Try to eliminate the subtraction signs from one set of factors by changing variables. Let $a = 2020 - \hat{a}$ and $b = 2017 - \hat{b}$. This can make it easier to find the values of $\hat{a}$ and $\hat{b}$, and additionally values of $\hat{a}$ and $\hat{b}$ will be small.

$$\begin{aligned}\frac{2020^2 - a^2}{2017^2 - b^2} &= \frac{2020^2 - (2020 - \hat{a})^2}{2017^2 - (2017 - \hat{b})^2} \\ &= \frac{\big(2020 + (2020 - \hat{a})\big)\big(2020 - (2020 - \hat{a})\big)}{\Big(2017 + (2017 - \hat{b}))(2017 - (2017 - \hat{b})\Big)} \\ &= \frac{(4040 - \hat{a})(\hat{a})}{(4034 - \hat{b})(\hat{b})}\end{aligned}$$

Notice that if $\hat{a} = 7$ and $\hat{b} = 1$, then we will get 4033 in both the numerator and denominator, which will cancel.

$$\begin{aligned}\frac{(4040 - 7)7}{(4034 - 1)1} &= \frac{4033 \times 7}{4033 \times 1} \\ &= \frac{7}{1} \\ &= 7.\checkmark\end{aligned}$$

We can prove that these are the only values of $\hat{a}$ and $\hat{b}$ that work; we know that $\hat{a} - \hat{b} = 6$ and $\hat{b}$ must be a divisor of $\hat{a}$, so there are only 2 possible ways, $(7, 1)$ and $(12, 6)$, and only one of them, $(7, 1)$, works.

Thus $\hat{a} = 7$ and $a = 2020 - \hat{a} = 2020 - 7 = \boxed{2013}$.

Earned 0 of 3 points. 0%

Super Bonus Questions

These questions are an extreme challenge! Some of the most interesting, creative and clever questions are the hardest ones. So try every single one. Even if you can't figure it out, you'll still be deepening your understanding of the important math concepts that underlie this course.

18 Find the sum of all three-digit integers such that 13 times the sum of their digits equals the number itself.

Your Answer	[no answer given]
Correct Answer	468

Explanation

Let's set some variables up.

$$\begin{aligned} H &= \text{hundreds digit} \\ T &= \text{tens digit} \\ U &= \text{units digit} \end{aligned}$$

Not all of H, T, or U can be zero. Now use the given information that this number equals 13 times the sum of its digits to write the following equation:

$$\begin{aligned} 100H + 10T + U &= 13(H + T + U) \\ (100 - 13)H &= (13 - 10)T + (13 - 1)U \\ 87H &= 3T + 12U \\ \frac{87H}{3} &= \frac{3T + 12U}{3} \\ 29H &= T + 4U \end{aligned}$$

The maximum value of $T + 4U$ is $9 + 4(9) = 45$, so $29H \leq 45$, which means the only possible value of H is 1. Let's first assign values to U, since $4U$ can be larger than T. If $U = 1, 2, 3$, or 4, then T will larger than a single-digit number, which we can't have.

$$\begin{aligned} 29 &= T + 4U \\ U &= 5 && \to T = 29 - 4(5) = 9 \\ U &= 6 && \to T = 29 - 4(6) = 5 \\ U &= 7 && \to T = 29 - 4(7) = 1 \end{aligned}$$

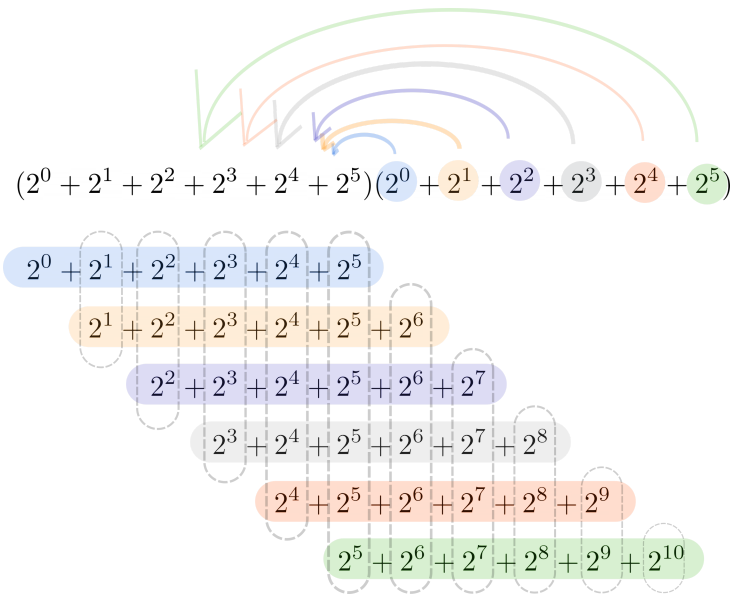
The three possible numbers are $117, 156$, and 195, which have a sum of $\boxed{468}$.

19 What is the value of
$2^0 + 2(2^1) + 3(2^2) + 4(2^3) + 5(2^4) + 6(2^5) + 5(2^6) + 4(2^7) + 3(2^8) + 2(2^9) + 2^{10}$?

✕	**Your Answer**	[no answer given]
	Correct Answer	3969

Explanation

First, imagine splitting up the powers of 2 and spreading them out. For example, in the term $5(2^4)$, split this into $2^4 + 2^4 + 2^4 + 2^4 + 2^4$. Next, line up the same powers of 2 in columns; the resulting shape will be a parallelogram, since there are more of the middle powers like 2^5 and fewer and fewer as we go towards smaller powers like 2^0 and 2^{10}, which each occur only once.



Each row has six terms, and each row has a common factor that makes it equal to $2^0 + 2^1 + 2^2 + 2^3 + 2^4 + 2^5$ multiplied to another power of 2. Thus we can do a little trick to factor this expression as the following:

$$\begin{aligned}
&2^0 + 2(2^1) + 3(2^2) + 4(2^3) + 5(2^4) + 6(2^5) + 5(2^6) + 4(2^7) + 3(2^8) + 2(2^9) + 2^{10} \\
&= (2^0 + 2^1 + 2^2 + 2^3 + 2^4 + 2^5)(2^0 + 2^1 + 2^2 + 2^3 + 2^4 + 2^5) \\
&= (2^6 - 1)(2^6 - 1) \qquad \text{(Using formula for sum of powers of 2)} \\
&= 2^{12} - 2(2^6) + 1 \\
&= 4096 - 128 + 1 \\
&= \boxed{3969}.
\end{aligned}$$

20 What is the value of $2^0 + 3(2^1) + 6(2^2) + 7(2^3) + 6(2^4) + 3(2^5) + 2^6$?

Your Answer	[no answer given]
Correct Answer	343

Explanation

One sure-fire approach is to calculate each term and add the terms up. A neat trick, however, is to see that this can be factored in the following way:

$$\begin{aligned}
&2^0 + 3(2^1) + 6(2^2) + 7(2^3) + 6(2^4) + 3(2^5) + 2^6 \\
&= (2^0 + 2^1 + 2^2)(2^0 + 2^1 + 2^2)(2^0 + 2^1 + 2^2) \\
&= (2^3 - 1)^3 \\
&= 7^3 \\
&= \boxed{343}.
\end{aligned}$$

Summary

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